

Database Design Project - Phase 2 - Conceptual / Logical Database Design [25 %]

Assign On: February 12th, 2026

Due On: February 27th, 2026

Step 5: EER modeling

Based on the Step 1 ~ 4 in the previous phase, in phase II draw an Enhanced Entity Relationship (EER) diagram for the problem you chose. This EER modeling should adhere to the principles taught in the course. It should also use the notation taught in the lectures and exercise labs. An Entity should have keys clearly identified and relationships should have the cardinality information. The resulting EER diagram should contain:

- At least 10 entities
- At least 10 relationships (1:1, 1:N, and N:M)
- Some attributes on relationships
- At least 1 Weak entity (you can add a weak entity on purpose if none exists)
- At least 1 Generalization/Specialization

Step 6: Creating Relations

Based on the EER diagram, you should create relations using the relational model in the lecture. Make sure to identify the primary key and foreign keys in all the relations.

- Identify other keys (candidate keys) in your relations
- Identify functional dependencies in your application and list them

Step 7: Relational Algebra (Chapter 8)

Formulate the queries in Step 3 using relational algebra. If, at this point, you are unable to state the queries in relational algebra, this is probably because your database is not powerful enough. Then, you must go back to step 2 and see how the database can be improved.

[Submission guideline for Phase II]

You should submit the full and legible report with a cover page that includes Team name, names of all members, the title of the project, and abstraction (brief description of the project). The body of Phase II documentation should include sufficient detail to describe all steps of Phase II:

- (Step5) ER diagram of the database
- (Step6) Relational database schema (including diagram)
- (Step7) Give the relational algebra expressions for the queries

In conclusion, describe what you have learned from the project, and what you need for the better project output.

The name of your submitted documentation file should be your "Project#_phase2". You can use your Group number for #. For example, "Group1_phase2.doc". Then, you will submit your file on Canvas.